

Personal Expense Tracker with Data Visualization

1. Introduction

Managing personal finances manually can be difficult and time-consuming. This project provides an automated solution for tracking expenses, generating reports, scanning receipts, and obtaining financial insights using Artificial Intelligence techniques.

The system provides a user-friendly interface that helps users monitor spending habits and make better financial decisions.

2. Objectives

- Track daily expenses efficiently.
- Store financial records digitally.
- Generate visual reports.
- Scan receipts automatically.
- Predict expense categories using Machine Learning.
- Analyze spending behavior.
- Generate AI-powered recommendations.
- To calculate total expenses automatically.
- To generate category-wise expense summaries.
- To improve financial awareness and budgeting.

3. Technologies Used

Technology	Purpose
Python	Core Programming Language
Pandas	Data Processing and Analysis
Matplotlib	Data Visualization
Scikit-Learn	Machine Learning
Tesseract OCR	Receipt Scanning
CSV File	Data Storage
VS Code	Development Environment
HTML, CSS	Frontend Structure and styling
Flask	Web Framework
CSV File Storage	Database

4. System Requirements

Hardware Requirements

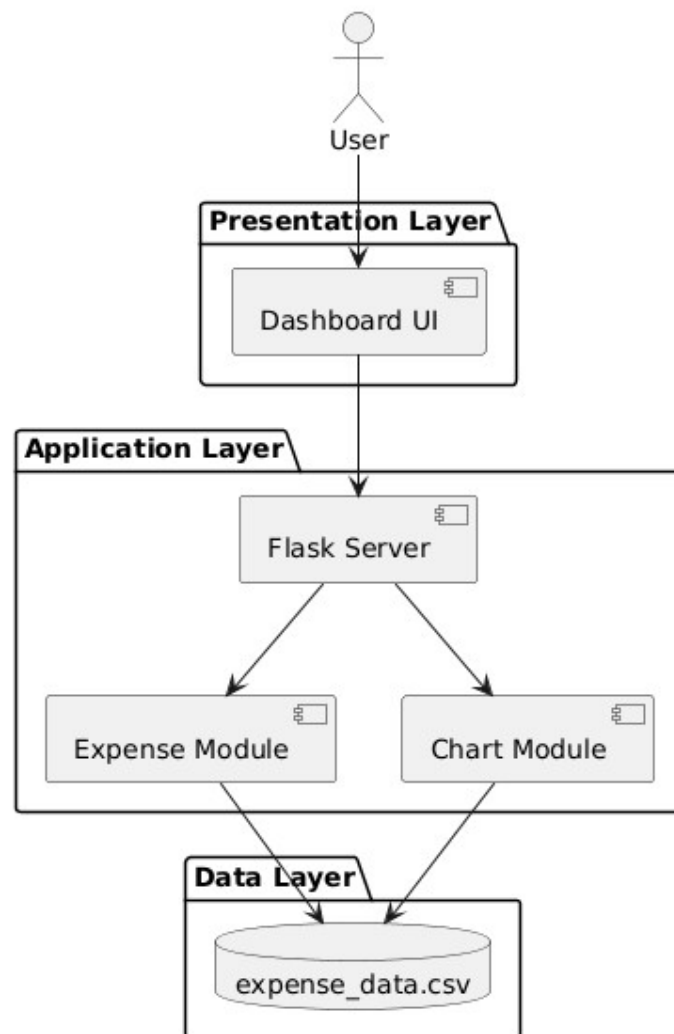
- Processor: Intel i3 or above
- RAM: 4 GB or above
- Storage: 500 MB free space

Software Requirements

- Python 3.x
- Pandas Library
- Matplotlib Library
- Tesseract OCR
- Visual Studio Code

5. System Architecture

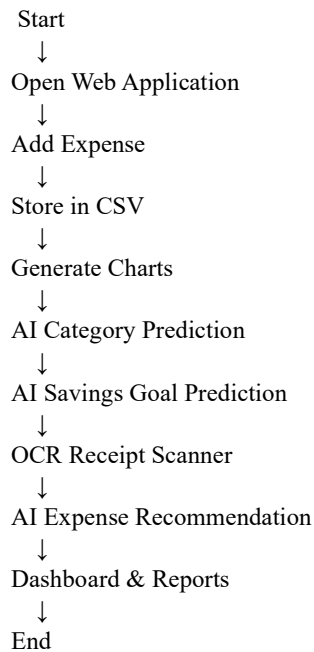
Personal Expense Tracker - Three Tier Architecture



6. Workflow

1. User enters expense details.
2. Expense data is stored in a CSV file.
3. Pandas reads and processes the data.
4. The system calculates total and category-wise expenses.
5. Matplotlib generates charts for visualization.
6. Results are displayed to the user.

Flow



7. Methodology

1 : Expense Data Collection

Users enter expense details such as date, category, amount, and description through the web interface

2 : Data Storage

The entered expense information is stored in a CSV file for further processing and analysis.

3 : Data Visualization

Expense data is analyzed and visualized using pie charts and bar charts to understand spending patterns.

4 : OCR Receipt Processing

Tesseract OCR extracts text and amount information from uploaded receipt images and automatically records expenses.

5 : Recommendation & Analysis

The system analyzes spending behavior and generates AI-based recommendations and savings predictions to improve financial planning.

8. MODULES AND MILESTONES

Milestone 1

1.1: EXPENSE MANAGEMENT MODULE

This module allows users to add, store, view, and manage expense records.

Features

- Add Expense
- View Expenses
- Calculate Total Spending
- Store Records in CSV

```
def add_expense():  
    date = input("Enter Date (YYYY-MM-DD): ")  
    category = input("Enter Category: ")  
    amount = float(input("Enter Amount: "))  
    description = input("Enter Description: ")  
  
    new_data = pd.DataFrame({  
        "Date": [date],  
        "Category": [category],  
        "Amount": [amount],  
        "Description": [description]  
    })  
  
    new_data.to_csv(FILE_NAME, mode='a', header=False, index=False)  
  
    print("Expense Added Successfully!")
```

EXPENSE

PERSONAL FINANCE TRACKER

TOTAL SPENT
₹0

NEW ENTRY

dd/mm/yyyy

Category

Amount

Note (optional)

Add →

Milestone 2

2.1 : DATA VISUALIZATION MODULE

This module converts expense data into visual charts for better understanding of spending patterns.

Features

- Expense Pie Chart
- Category Expense Bar Chart



2.2: Expense Storage

Features:

- Save records to CSV.
- Maintain expense history.

```
expense_data.csv
1 Date,Category,Amount,Note
2 2026-05-02,Food,1230,
3 2026-06-05,Rent,15000,
4 |
```

Milestone 3

3.1: AI & MACHINE LEARNING MODULE

This module uses Machine Learning to predict expense categories and generate intelligent recommendations.

AI Model Used

Multinomial Naive Bayes Classifier

Workflow

Expense Description



Text Processing



Count Vectorization



Naive Bayes Model



Predicted Category

- AI CATEGORY PREDICTION

Predict →

PREDICTED CATEGORY

Food

- AI SAVINGS GOAL PREDICTOR

Goal name

Target amount (₹

Monthly saving (

Predict →

Laptop to buy

TARGET

₹50000

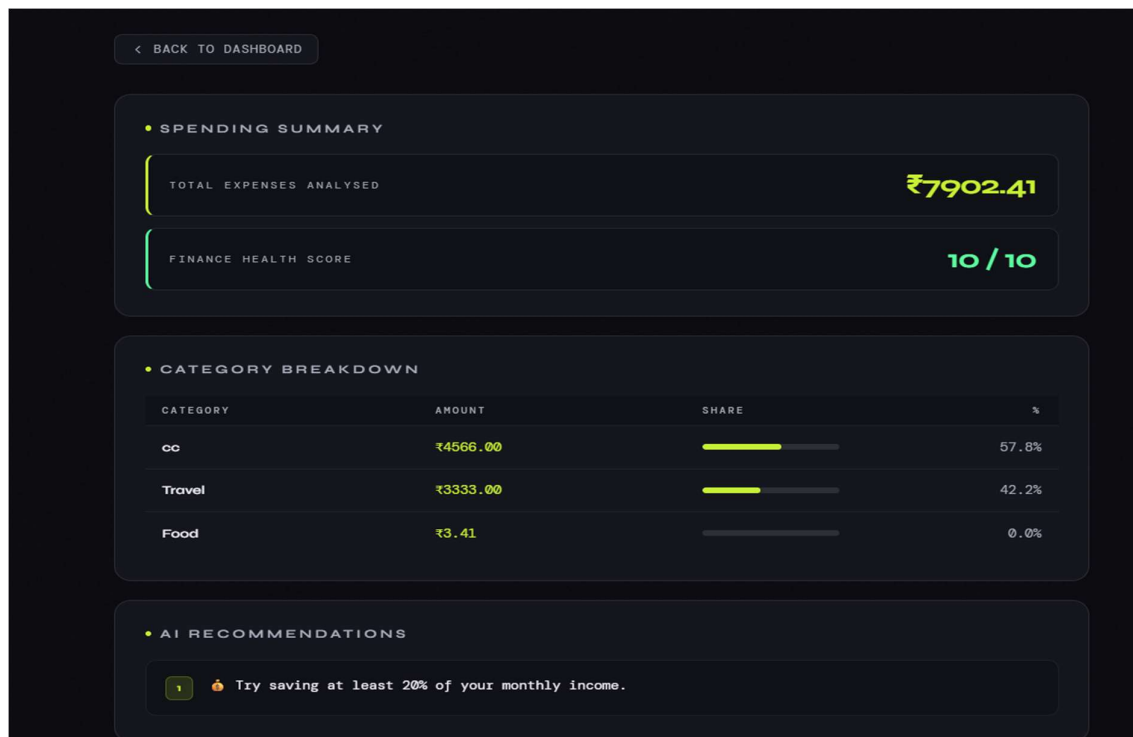
MONTHLY SAVING

₹3000

ESTIMATED TIME

23mo

🔥 Goal achievable within 2 years.



Milestone 4

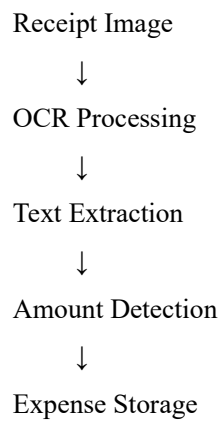
4.1: OCR RECEIPT SCANNER MODULE

The OCR module extracts text and expense information from uploaded receipts automatically.

OCR Technology

Tesseract OCR

Workflow



9. Models Used

1. Multinomial Naive Bayes
Purpose :
Expense Category Prediction

2. Tesseract OCR

Puurpose :

Receipt Text Extraction

Receipt Image



Extract Text



Identify Total Amount



Store Expense

3. Recommendation Engine

Purpose:

Analyze spending patterns and generate savings suggestions.

10.Future Enhancements

- SQLite/MySQL database integration.
- Monthly PDF Reports
- Budget alerts and notifications
- Cloud storage support.

11.Conclusion

The AI-Powered Personal Expense Tracker successfully integrates Expense Management, Data Visualization, Machine Learning, OCR, and Recommendation Systems into a single platform. The project demonstrates practical implementation of Artificial Intelligence and Python technologies while providing meaningful financial insights to users.

The system can be further enhanced with cloud deployment, database integration, and advanced forecasting models.